Remote-Sensing Interpretation for Soil Elements Using Adaptive Feature Fusion Network

Yue Lu, Kang He, Haoran Xu, Yusen Dong, Wei Han, *Member, IEEE*, Lizhe Wang, *Fellow, IEEE*, and Dong Liang

***Abstract*— Soil elements refer to different types of soils with unique colors, textures, and particle sizes. Their interpretation is essential for agriculture, ecological environment, and land permeability assessment. This typically requires experts with dual knowledge of geology and remote sensing. With the increasing volume of remote-sensing data, the traditional "visual interpretation" and "field survey" technique is no longer sufficient to meet the demands. Because of the challenges such as the fine structure of soil, complex and variable natural scenes, and strong spatial variability, there remains a considerable gap between the accuracy of deep-learning-based methods and expert interpretation. To improve the accuracy of intelligent soil elements interpretation, this study proposes a soil interpretation framework coupling implicit knowledge with a multispectral image (SIFCIM). This framework quantifies implicit knowledge (IK), such as interpretation symbol and terrain feature, into matrix data [interpretation symbol distance (ISD) field and digital elevation model (DEM)]. To align with the SIFCIM, an IK-guided adaptive feature fusion network (IAFFNet) is constructed, which enhances the utilization efficiency of auxiliary features through an adaptive implicit feature fusion (AIF) module and a global feature dependence (GFD) module. Experimental results demonstrate that IAFFNet outperforms interpretation methods with a single remote-sensing image, achieving approximately 4.34% and 6.62% improvements in overall pixel accuracy (oPA) and mean intersection over union (mIoU), respectively. These results validate the effectiveness and robustness of the IK-guided approach in soil elements interpretation. To our knowledge, this work is the first to apply the concept of IK to soil elements interpretation, providing a novel insight for related research.**

***Index Terms*— Deep learning, feature fusion, implicit knowledge (IK), soil elements interpretation.**

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Yue Lu is with the International Research Center of Big Data for Sustainable Development Goals, Beijing 100094, China, and also with the Key Laboratory of Geological Survey and Evaluation, Ministry of Education, China University of Geosciences, Wuhan 430074, China (e-mail: luyue@cug.edu.cn).

Kang He is with the Key Laboratory of Geological Survey and Evaluation, Ministry of Education, China University of Geosciences, Wuhan 430074, China.

Haoran Xu, Yusen Dong, and Wei Han are with the School of Computer Science and the Hubei Key Laboratory of Intelligent Geo-Information Processing, China University of Geosciences, Wuhan 430078, China.

Lizhe Wang is with the International Research Center of Big Data for Sustainable Development Goals, Beijing 100094, China, and also with the School of Computer Science and the Hubei Key Laboratory of Intelligent Geo-Information Processing, China University of Geosciences, Wuhan 430078, China.

Dong Liang is with the International Research Center of Big Data for Sustainable Development Goals, Beijing 100094, China, and also with the Key Laboratory of Digital Earth Science, Aerospace Information Research Institute, Chinese Academy of Sciences, Beijing 100094, China (e-mail: liangdong@radi.ac.cn).

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I. Introduction

SOIL is a vital component of the Earth's terrestrial ecosystem [1], playing a crucial role in various environmental processes and supporting a wide range of human activities [2]. Soil elements refer to different types of soils with distinct characteristics such as unique colors, textures, and particle sizes [3]. The formation of soil elements is influenced by various factors including parent material, climate, terrain, and vegetation cover [4]. Over time, these factors interact with each other, resulting in unique physical, chemical, and biological properties [4], [5], [6].

Accurate interpretation and mapping of soil elements are of significant importance for land managers and decision-makers [7] in carrying out tasks such as agricultural planning, ecological environment monitoring, land permeability assessment, and soil bearing capacity analysis [8], [9], [10], [11]. However, traditional methods for interpreting soil elements primarily rely on "visual interpretation" and "field surveys" [12], which not only require experts with dual knowledge in geology and remote sensing, but also necessitate extensive ground surveys, soil sampling, and laboratory analysis [13], [14], [15]. These approaches are labor-intensive and have low production efficiency.

With the exponential growth in remote-sensing image data volume and the increasing richness of data types [16], coupled with the advancement of deep-learning techniques, overcoming these limitations is becoming possible [17]. Addressing the key scientific and technological challenge of how to leverage deep learning for the rapid extraction and intelligent interpretation of soil elements from remote-sensing images has become an urgent breakthrough that awaitscite [18].

In recent years, deep-learning semantic segmentation algorithms have emerged as important tools for intelligent interpretation of remote-sensing images due to their ability to achieve fine-grained reasoning by performing dense pixel-wise prediction and inference labeling [19]. For instance, Zhan et al. [20] introduced the vegetation-feature-sensitive

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focus perception module and the adaptive context inference model to enhance vegetation object feature extraction. Liu et al. [21] introduced three CNN algorithms to extract lithological features in their hyperspectral lithology classification work, which significantly improved the classification accuracy.

However, because of the fine structure of soil, complex, and variable natural scenes, and strong spatial variability, the application of semantic segmentation methods solely based on remote-sensing images faces several challenges, such as limited interpretable elements, inaccurate element boundaries, and unreliable attribute reliability [22]. The technical barriers of existing methods can be attributed to the following reasons.

1) *Inadequate knowledge application:* The interpretation process lacks operational rule constraints and expert knowledge assistance [23]. However, geological rules and expert knowledge summarize and generalize knowledge from geology, geography, and extensive experiential knowledge. The challenge lies in how to enable deep networks to use such abstracted and fragmented knowledge to improve the accuracy of intelligent soil elements interpretation [24].
2) *Insufficient algorithmic inference and self-learning:* The lack of tight integration between interpretation model algorithms and application requirements results in low algorithmic accuracy and poor scene adaptability [18]. This hinders the support for intelligent interpretation of high-resolution remote sensing in complex natural scenes. The spatial distribution of soil is not only influenced by many natural environments such as topography, landforms, and weather, but also by the transport and accumulation of parent material during soil formation processes [25]. The former leads to a random spatial distribution of soil with high interclass similarity and intraclass dissimilarity [26]. The latter leads to a distribution of soil classes according to a certain pattern, known as structural characteristics [27]. The random and structural characteristics of soil spatial distribution place requirements on the establishment of interpretation model algorithms.

To address the above challenges, remote-sensing intelligent interpretation methods coupled with auxiliary data have been proposed [28], [29], [30]. Sherrah [28] proposed a downsampling-free hybrid network to obtain high-precision semantic annotation results by mixing high-resolution aerial image features and a digital surface model (DSM). Song et al. [29] constructed a deep hashing-based hyperspectral and LiDAR feature fusion framework to enrich semantic features as a way to improve land cover interpretation. Zhou et al. [30] performed cross-layer feature aggregation and cross-modal feature fusion to filter redundant features and complement high-resolution remote-sensing images and DSMs.

Furthermore, knowledge graphs have been applied by researchers in intelligent remote-sensing interpretation tasks due to their ability to consider rich connections among different scene elements [31], [32], [33]. Li et al. [31] introduced a deep alignment network that leverages knowledge graphs to achieve zero-shot and generalized zero-shot scene classification for remote-sensing images. Cui et al. [32] proposed a parallel walking algorithm-based method for matching remote-sensing scene graphs with knowledge graphs to consider the visual features of complex scenes. Gun and Chen [33] utilized knowledge graphs and an optimal segmentation algorithm to address misclassification caused by neglecting spatial and contextual relationships between segmented objects in high-resolution remote-sensing image classification.

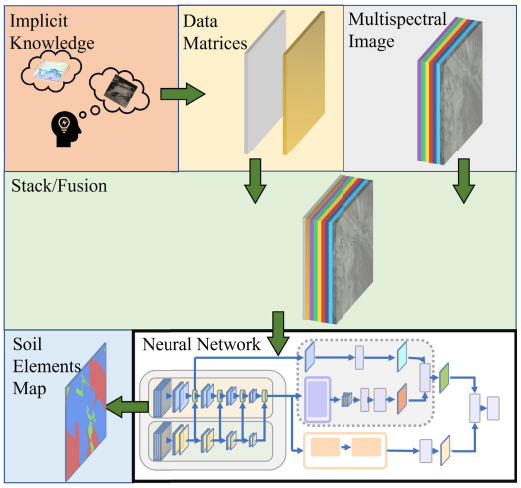


Fig. 1. SIFCIM quantifies IK into data matrices and stacks or fuses it with a multispectral image. The coupled data will be inputted to a neural network for soil elements interpretation.

However, the auxiliary data methods often rely on data derived from remote sensing [34], and knowledge graph methods require extensive and complex geological data and domain expertise, posing high demands on researchers' knowledge reservoir and the representation and modeling methods of knowledge graphs [35]. This study proposes a soil interpretation framework coupling implicit knowledge with a multispectral image (SIFCIM) shown in Fig. 1. Implicit knowledge (IK) refers to knowledge that is not explicitly stated or communicated, but is understood by individuals with a shared context or background. It is an important aspect of human cognition and decision-making [36]. Interpretation symbols and terrain features extensively used in geological interpretation are examples of IK. In SIFCIM, to facilitate neural network processing, these IK elements are quantified into data matrices such as interpretation symbol distance (ISD) fields and digital elevation models (DEMs).

To align with the SIFCIM framework and fully leverage its advantages, this study develops an IK-guided adaptive feature fusion network (IAFFNet). IAFFNet consists of a dual-branch structure adaptive implicit feature fusion (AIF) backbone network, a multiscale feature extraction (MSFE) module, and a global feature dependence (GFD) module. AIF is responsible for extracting and coupling features from MSI and IK. The coupled features are then fed into the MSFE for multiscale feature extraction and the GFD for establishing

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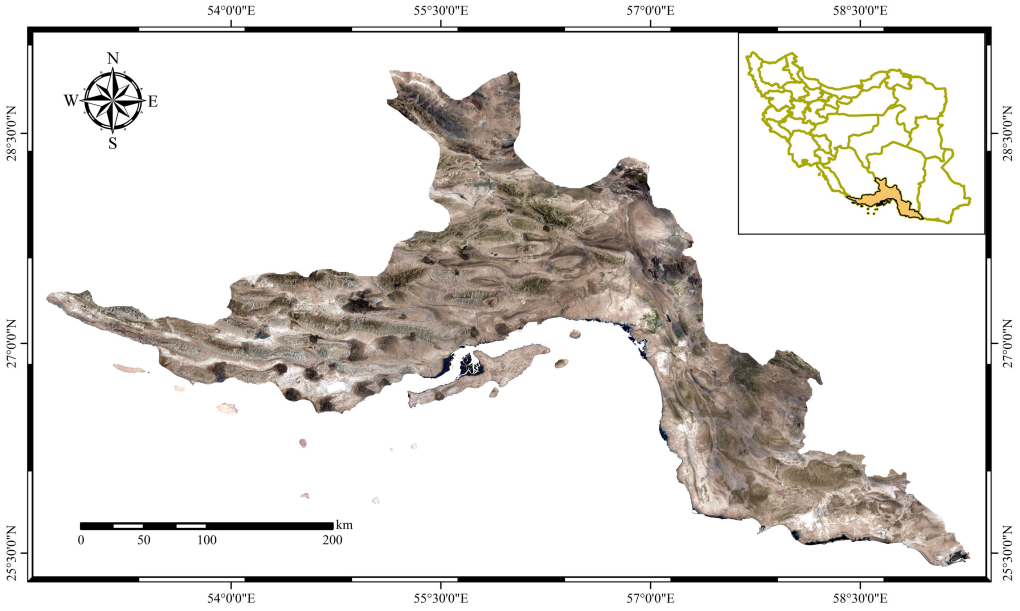


Fig. 2. RGB image of Hormozgan. The region is located in southern Iran, adjacent to the Persian Gulf and the Strait of Hormuz, characterized by mountainous areas.

global correlations of soil features. The contributions of this study are mainly in three aspects.

1) This study investigates the feasibility of using IK to improve the low accuracy of soil elements interpretation caused by insufficient algorithmic inference, self-learning, and inadequate knowledge application.
2) The feasibility and generality of the SIFCIM are explored. By coupling IK with MSI, fragmented expert knowledge can be conveniently and efficiently applied to intelligent soil elements interpretation. This offers new insights for related work.
3) This study proposes and validates the effectiveness and superiority of the IAFFNet. Multiple experimental results demonstrate that IAFFNet achieves significantly higher interpretation accuracy compared to methods without knowledge coupling, with overall pixel accuracy (oPA) and mean intersection over union (mIoU) improved by approximately 4.34% and 6.62%, respectively.

II. Research Area and Dataset

A. Research Area

The study area is located in Hormozgan Province in southern Iran, adjacent to the Persian Gulf and the Strait of Hormuz, as shown in Fig. 2. Because of the proximity of the Zagros Mountains and the Kuh-e Kuhroud Mountains, the region consists mostly of mountainous areas, with small alluvial plains along the southern coast of the Strait of Hormuz. Influenced by a temperate continental climate and a highland mountain climate, the area exhibits a wide variety of soil elements [37]. The dominant rock types are Quaternary volcanic rocks and Mesozoic greenschist rocks, with the predominant landforms consisting of gravel soil and rock formations, along with some fine-grained soil and a small amount of sandy soil. Soil salinity increases gradually from the rocky hills to the beach surface. The topography and climate play crucial roles in the soil formation processes in this region [38]. The study area is located near the sea and coastline, and the identification of the material composition and genetic types of the area is extremely important for the preparation of territorial spatial planning, the overall protection of the geological environment, systematic restoration, and comprehensive management.

B. Multimodal Dataset

Soil elements interpretation is the fundamental task in regional geological survey mapping. It involves determining soil properties based on the spectral information, spatial characteristics, and expert knowledge derived from remote-sensing images. In this study, a multimodal dataset was created for soil elements interpretation using Sentinel-2 multispectral image and IK, which includes ISD and DEM. The dataset comprises 2746 nonrepetitive regions, each associated with MSI, ISD, DEM, and Label data, with a dimension of 224 × 224 pixels.

1) *Sentinel-2:* This is a bipolar orbiting imaging satellite with 13 spectral bands. In this study, three bands with a spatial resolution of 60 m, which provide limited value, were excluded [40]. The remaining ten bands were selected as MSI inputs (see Table I). Among them, the bands with a spatial resolution of 20 m were resampled to 10 m. These ten bands contain key information relevant to soil elements interpretation, such as water quality assessment (Band 2), vegetation coverage (Band 3), land use (Band 4), vegetation health status (Bands 6-8A), and soil moisture (Bands 5-8A).
2) *DEM:* Terrain plays a key role in the soil formation process, influencing the redistribution of soil moisture, heat, and parent material. In this study, the DEM is

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TABLE I

MAIN BANDS OF THE SENTINEL-2 USED IN THIS STUDY [39]

Spatial Resolution (nm)	Bands	Central wavelength (nm)
10	Band 2-Blue	490
	Band 3-Green	560
	Band 4-Red	665
	Band 8-NIR	842
20	Band 5-Red edge	705
	Band 6-Red edge	740
	Band 7-Red edge	783
	Band 8A-Narrow NIR	865
	Band 11-SWIR	1610
	Band 12-SWIR	2190

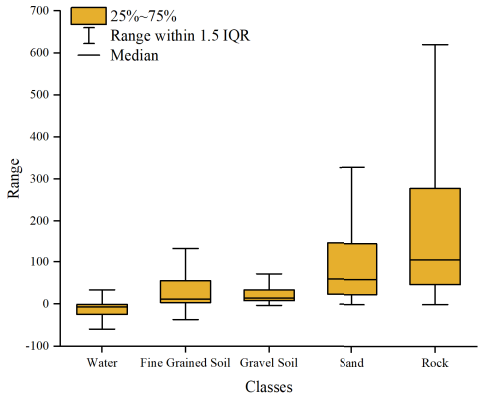


Fig. 3. Distribution on DEM of different land cover types.

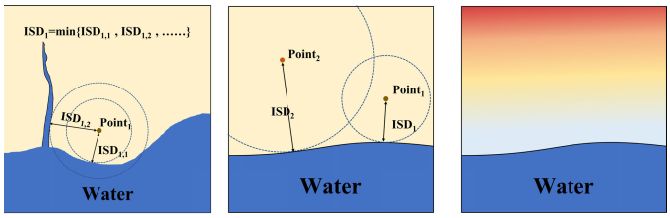


Fig. 4. ISD field is generated by calculating the minimum Euclidean distance from each pixel to the water bodies, using water bodies as interpretive symbols. (a) Take the minimum Euclidean distance from Point_1 to the water body as the ISD_1 of the point. (b) Obtain ISD_1 and ISD_2 according to the method in (a). (c) Solve the ISD for any point to generate the ISD field.

introduced as a quantification of the terrain for soil elements interpretation. The original DEM data is derived from the ASTER satellite data resources, with a spatial resolution of 30 m, vertical accuracy of approximately 20 m, and horizontal accuracy of approximately 30 m [41]. Batch processing tools in ArcGIS were employed to perform data format conversion, mosaicking, clipping, and resampling, resulting in a 10-m resolution DEM dataset. Fig. 3 shows the distribution of the different elements on the DEM. It can be observed that various soil attributes exhibit significant variations in elevation, which is a valuable characteristic for soil elements interpretation.

3) *ISD Field:* The study area is located near the coast and shoreline. Quaternary alluvium deposits, slope deposits, and flood deposits in the area were formed by the transportation of flowing water bodies. Therefore, water bodies serve as crucial interpretation symbols for experts when interpreting soil elements. To input the influence of interpretation symbols into the neural network, this study quantifies them in the form of data matrices. As shown in Fig. 4, for any point within the research area excluding water bodies, the minimum Euclidean distance from that point to the edge of the water body is calculated as the ISD for that point. The ISD of all points collectively forms the ISD of the research area. This innovative approach transforms the implicit influence of interpretation symbols into data matrices, laying the foundation for coupling interpretation symbols with remote-sensing images.

TABLE II

PROPORTION OF CLASSES IN THE TRAIN DATASET AND TEST DATASET

Class	Proportion	
	Train	Test
Water	0.193421518	0.199889734
Fine-Grained Soil	0.120857904	0.125329987
Gravel Soil	0.286075242	0.290443038
Sandy Soil	0.024797712	0.025135499
Rock	0.374847623	0.359201741

C. Labels

The labels are shown in Fig. 5. The label dataset is the result of manual interpretation by senior geologists and includes five categories (0—water, 1—fine-grained soil, 2—gravel soil, 3—sandy soil, 4—rock). The gravel soil, sandy soil, and fine-grained soil are classified according to their grain size (2–60, 0.075–2, and 0–0.075 mm, respectively) [42]. As shown in Table II, the distribution of the five categories in the training and test samples is roughly similar, and the proportion of each category is not balanced. Among them, the rock category accounts for the largest proportion at over 35%; the water body accounts for a moderate proportion, meeting the requirement of being a marker for interpretation. Of the soil elements, gravel soil accounts for the largest proportion at over 28.5%. This is followed by fine-grained soil, with a proportion of around 12%. Sandy soil only accounts for about 2.5%, which poses a challenge for sandy soil elements interpretation. To enable the model to learn the sandy soil characteristics better and counteract the negative effects of the class imbalance, a hard-to-score sample mining method is introduced to facilitate the comparison of the realistic performance of each model in the model experiments [43].

III. METHODOLOGY

Expert knowledge, for example, topography and interpretation symbol, is a summary generalization of a large amount of empirical knowledge. To facilitate the utilization of this knowledge by deep-learning networks in soil elements

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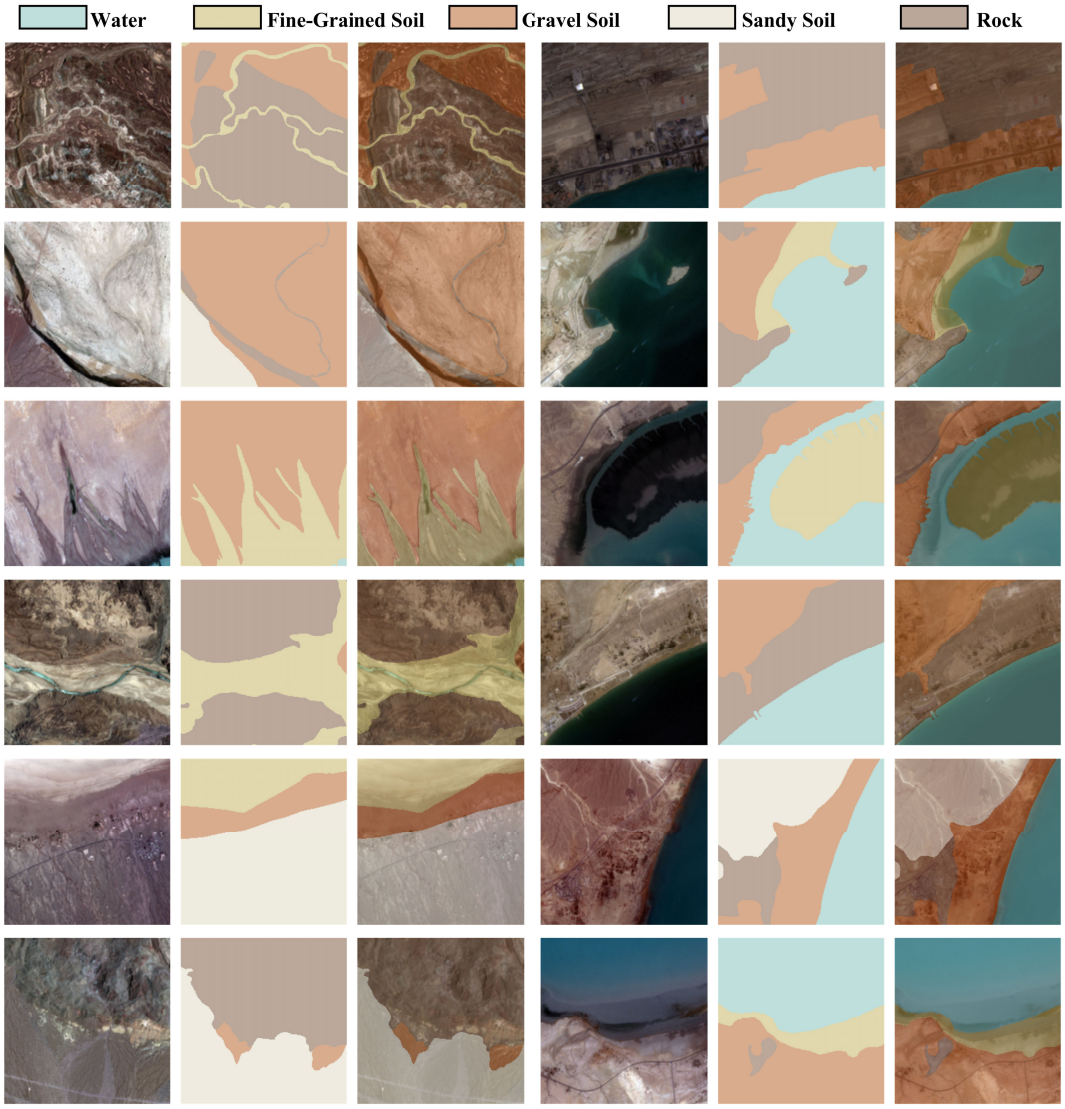


Fig. 5. Example of images, labels, and label mask visualizations.

interpretation tasks more easily and effectively, we propose the SIFCIM framework, which allows IK to guide the network. SIFCIM quantifies IK in the form of data matrices and combines them with multispectral images as input to the neural network. Using the idea of multisource fusion, the IK is allowed to fuse with the multispectral images to guide the network for the interpretation of soil elements.

To align more closely with the principles of SIFCIM, this study has designed IAFFNet. Fig. 6 shows an overview of the architecture of IAFFNet and how the data flows, including the following three main points.

1) *AIF:* AIF consists of a multispectral image branch (MSIB) and an IK-guided branch (IKGB) that extracts features from each of the two modalities. The features are then fused by the attention feature complementary (AFC) module [44], where the IK features can be utilized more effectively.
2) *MSFE:* Because soil elements have different distributions at different spatial scales, MSFE introduces atrous spatial pyramid pooling (ASPP) [45] to extract multiscale feature information. While retaining the multiscale features of soil elements, they are further fused with low-level features to recover the lost texture and boundary information.
3) *GFD:* IAFFNet introduces GFD for the spatially distributed structural features of the soil. The GFD contains two criss-cross attention (CCA) modules [46] that share parameters and are responsible for establishing remote dependencies between features one by one, thus achieving global feature association.

A. IK Quantification

The DEM data used to represent the terrain is directly accessible. Therefore, the focus is on the quantification method for interpretation symbols. In this study, the minimum Euclidean distance between pixels and interpretation symbols is employed as a digital representation of the influence of interpretation symbols on soil elements interpretation, which

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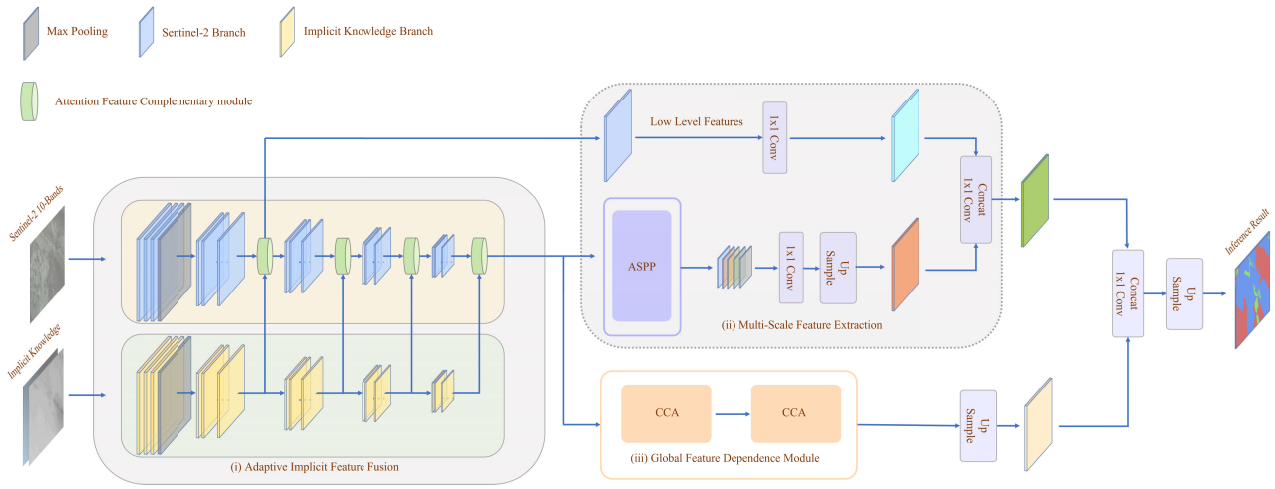


Fig. 6. Overview of the proposed IAFFNet for the interpretation of soil elements. (i) Dual-branch backbone consisting of two ResNet-50 models, utilizing AFC to fuse MSI and IK features. (ii) MSFE module, extracting multiscale information from the output of (i) and completing low-level features. (iii) Global dependencies for the features from (i).

can be expressed as follows:

$$\mathrm{ISD}_{A,I} = \min\left\{\sqrt{(x_A - x_I)^2 + (y_A - y_I)^2}\right\} \tag{1}$$

where $\mathrm{ISD}_{A,I} \in \mathbf{ISD}$ denotes the minimum Euclidean distance from any point $A(x_A, y_A) \in \mathbf{S}$ in the study area to the interpretation symbol $I(x_I, y_I) \in \mathbf{I}$ [47]. The calculated **ISD** is used as one of the inputs to the following network.

B. Adaptive Implicit Feature Fusion

AIF is a two-branch structure that allows MSI and IK to be fed into the MSIB and the IKGB, fusing the two modal features after each layer via the AFC module [44], enabling efficient and targeted feature extraction.

The specific flow of the AFC module is illustrated in Fig. 7. First, the features S from MSIB and I from IKGB are fed into the global average pooling layer, which is used as a descriptor to represent the channel-level features; then, a 1×1 convolution layer is used to extract the correlation between channels; then, a sigmoid function is used for activation to obtain the weights of each channel (ranging from 0 to 1); by calculating the outer product of the input features and channel weights of the two branches to efficiently learn the correlation between channels and targeted learning of certain channels; finally, the output of the two branches is summed term by term to obtain the features $M \in \mathbb{R}^{C\times H\times W}$ complemented by the IK features. This is shown as follows:

$$M = S \otimes \Theta[\Psi_\Delta(S)] + I \otimes \Theta[\Psi_\Theta(I)] \tag{2}$$

where $\Psi(\cdot)$ represents global average pooling and 1×1 convolution, $\Theta(\cdot)$ represents the Sigmoid function, and $\otimes$ represents the inner product. For an MSIB feature $S = [s_1, s_2, \ldots, s_C] \in \mathbb{R}^{C\times H\times W}$, its channel-level feature $Y \in \mathbb{R}^C$ can be generated by global average pooling. For a channel d in S, its feature $E_{d,\mathrm{ave}} \in Y$ can be computed as follows:

$$E_{d,\mathrm{ave}} = \frac{1}{H \times W} \sum_{h=0}^{H-1} \sum_{w=0}^{W-1} d_{h,w}. \tag{3}$$

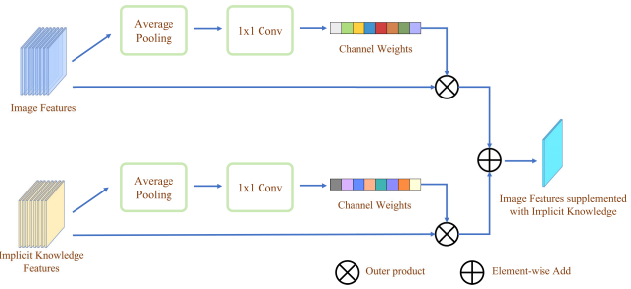


Fig. 7. Details of the AFC module. For each layer of extracted MSI and IK features, global average pooling and 1×1 convolution are employed to obtain channel weights. AFC is utilized to fuse the weighted features, and the resulting fusion is used as the input for the next layer of MSIB.

A similar approach can be used to extract IKGB channel-level features $I = [i_1, i_2, \ldots, i_C] \in \mathbb{R}^{C\times H\times W}$. The introduction of this module in IAFFNet allows for higher-value channel weights to be obtained from features containing rich information, hence allowing for more efficient and rational extraction of the distribution patterns of the soil elements contained in IK features.

C. Multiscale Feature Extraction

Because soil elements have different distribution characteristics at different spatial scales, the MSFE is introduced in IAFFNet. The core of MSFE is ASPP [45], which uses convolution kernels with different expansion factors in ASPP to extract multiscale feature information. For example, Fig. 8 shows the comparison of the perceptual fields of the 3×3 dilated convolution with the different expansion factors r. As in Fig. 9, ASPP consists of a 1×1 convolution, three 3×3 null convolutions with expansion factors of 6, 12, and 24, respectively, and a global average pooling. Here, 1×1 convolution is used to extract the effective information of each channel; dilated convolution is used to extract the features of different receptive fields; global average pooling is used to extract the image-level features and is calculated in the same way as in (2).

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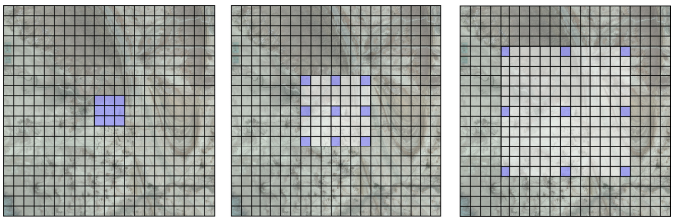


Fig. 8. Comparison of 3 × 3 dilated conv with $r = 1$, $r = 3$, and $r = 6$.

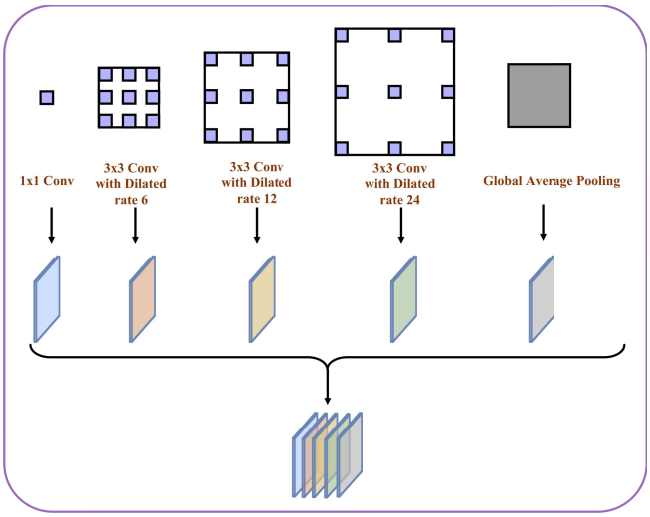


Fig. 9. Details of ASPP.

Dilated convolution allows the convolution kernel to have a wide range of receptive fields by setting different expansion factors r, so its mapping method differs from that of standard convolution. For an input feature $X \in \mathbb{R}^{C\times H\times W}$, the corresponding output Y_i for each of its positions i can be calculated as follows:

$$Y_i = \sum_k x_{i+r\cdot k} \cdot W_k. \tag{4}$$

To recover the textures and boundary features lost in high-level features, the multiscale features are fused with the low-level features LF extracted from the first layer in the AIF. After inputting a feature map $X \in \mathbb{R}^{C\times H\times W}$ obtained from the AIF into the MSFE module, the process of obtaining $Y \in \mathbb{R}^{C\times H\times W}$ containing the multiscale features is calculated as follows:

$$\begin{aligned} Y = \text{Concat}\{\text{Bi}[\text{Concat}[\text{Conv}_{1\times1}(X); \text{AC}_6(X); \\ \text{AC}_{12}(X); \text{AC}_{24}(X); \text{GAP}(X)]]; \text{LF}\} \end{aligned} \tag{5}$$

where $\text{Conv}_{1\times1}$ is 1 × 1 convolution, $\text{AC}_r(\cdot)$ is 3 × 3 dilated convolution with a dilated rate of r, $\text{GAP}(\cdot)$ is global average pooling, $\text{Bi}(\cdot)$ is quadratic linear interpolation, and $\text{Concat}(\cdot)$ is the concatenation function.

D. Global Feature Dependence

Because of the interdependencies of the distribution characteristics of the soil elements and their influence on each other, the GFD is added. The core of GFD is the CCA of two shared parameters, as shown in Fig. 10. CCA is a nonlocal method that treats each feature as a node and establishes a correlation between that node and all other feature nodes in its row/column to establish long-range dependencies in the horizontal or vertical direction.

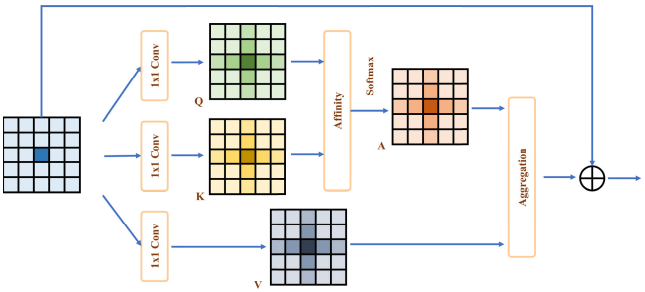


Fig. 10. Details of the CCA module.

For an input feature $X \in \mathbb{R}^{C\times H\times W}$, CCA first generates two feature maps $Q \in \mathbb{R}^{C'\times H\times W}$ and $K \in \mathbb{R}^{C'\times H\times W}$ and an adaptive matrix $V \in \mathbb{R}^{C\times H\times W}$ through three 1 × 1 convolutions. We can obtain a vector $Q_{\mathbf{u}} \in \mathbb{R}^{C'}$ that represents the features contained in a position of the feature map Q in the spatial dimension. Correspondingly, we can extract the feature vector of the row or column in which position $\mathbf{u}$ is located from K, thus obtaining a set $\Upsilon_{\mathbf{u}} \in \mathbb{R}^{C'\times(H+W-\Delta)}$. $\Upsilon_{i,\mathbf{u}}$ represents the ith element in $\Upsilon_{\mathbf{u}}$. The affinity operation is then performed on Q and K to obtain the attention graph $A \in \mathbb{R}^{(H+W-1)\times(W\times H)}$. The definition of the affinity operation is shown as follows:

$$d_{i,\mathbf{u}} = Q_{\mathbf{u}} \cdot \Upsilon_{i,\mathbf{u}}^T \tag{6}$$

where $d_{i,\mathbf{u}} \in D$ is the correlation measure between features $Q_{\mathbf{u}}$ and $Y_{i,\mathbf{u}}$, where $i = [1, \ldots, H + W - 1]$ and $D \in \mathbb{R}^{(H+W-1)\times(W\times H)}$. The attention graph A is obtained by activating D in the channel dimension through the softmax layer.

Similarly, we can obtain the feature vector $V_{\mathbf{u}} \in \mathbb{R}^C$ for any position $\mathbf{u}$ in V and the set of feature vectors $\Lambda_{\mathbf{u}} \in \mathbb{R}^{C\times(H+W-\Delta)}$ for the row or column in which this position is located. The aggregation operation is used to combine the input features X and the contextual information in order to obtain a feature X' containing the global feature dependencies. Aggregation is calculated as follows:

$$X'_{\mathbf{u}} = \sum_{i=0}^{H+W-1} A_{i,\mathbf{u}} \cdot \Lambda_{i,\mathbf{u}} + X_{\mathbf{u}} \tag{7}$$

where $X'_{\mathbf{u}}$ is the eigenvector of X' at position $\mathbf{u}$ and $A_{i,\mathbf{u}}$ is a scalar value of the attention map A at the corresponding position $\mathbf{u}$ in the ith channel.

Repeating the above process allows any location in X to obtain information about all other nodes, that is, establishing a global remote dependency between feature nodes. Fig. 11 shows the transfer of feature information from position η to position $\mathbf{u}$ after two CAA module calculations.

IV. Experiments and Analysis

A. Experimental Environment and Parameter Settings

In this study, we used the dataset presented in Section II, where the training, validation, and test sets have been randomly partitioned. Some classical semantic segmentation

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TABLE III
RESULTS OF ABLATION STUDY ABOUT IAFFNET

Method	Sentinel-2	DEM	ISD	AIF	GFD	oPA (%)	mIoU (%)
DeepLab v3+	✓	-	-	-	-	74.13	50.53
	✓	✓	-	-	-	76.42	52.77
	✓	-	✓	-	-	77.54	55.33
	✓	✓	✓	-	-	78.21	55.46
IAFFNet	✓	✓	-	✓	-	81.48	58.99
	✓	-	✓	✓	-	80.91	59.84
	✓	✓	✓	✓	-	81.44	60.17
	✓	✓	-	✓	✓	82.25	60.62
	✓	-	✓	✓	✓	80.46	59.86
	✓	✓	✓	✓	✓	82.55	62.08

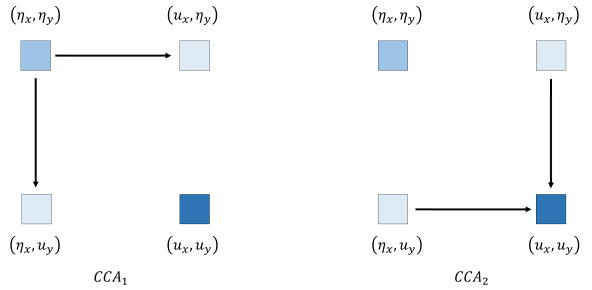


Fig. 11. Transmission path of information from position η to position $\mathbf{u}$.

networks were chosen for comparison: for example, fully convolutional networks (FCNs) [48], CCNet [46], and DeepLab v3+ [45], where the backbone of FCN uses ResNet-101, while the other two use ResNet-50. In addition, some of the latest models were also used as comparison objects in the experiments, such as Segformer [49], vision transformer (ViT) [50], bilateral network with guided aggregation (BiSeNet v2) [51], and bidirectional encoder representation from image transformers (BEiTs) [43]. Except for DeepLab v3+, all the above models involved in the comparison were implemented by MMSegmentation [52]. All of the above models use a simple stacking approach to incorporate IK. There are four types of input data: Sentinel-2 (SE), Sentinel-2 and DEM (SD), Sentinel-2 and ISD (SI), and Sentinel-2 and DEM and ISD (SDI). The experiments were conducted on a workstation equipped with an Intel i7-11700k CPU and an NVIDIA RTX 3080 (10G) GPU. Regarding the hyperparameter settings, the experiments set the batch size to 8 and the initial learning rate to 0.0001. All experiments used Adam's algorithm as the optimizer, and the overall experimental process was 150 epochs.

B. Evaluation Metrics

This work has used two metrics popular in semantic segmentation to evaluate the segmentation accuracy of the model, namely pixel accuracy (PA) and intersection over union (IoU).

PA is the ratio of the number of correctly classified pixels calculated to the number of all pixels in the class; it is the simplest metric to evaluate. The oPA refers to the calculation of the proportion of all correctly classified pixel points out of the total number of pixel points. This is shown in (8), where C refers to the number of categories and p_{ij} refers to the number of pixel points with actual category i and predicted to be j [53]

$$\text{oPA} = \frac{\sum_{i=0}^{C} p_{ii}}{\sum_{i=0}^{C} \sum_{j=0}^{C} p_{ij}}. \tag{8}$$

IoU is defined as shown in (9), where A is the true value map for a category and B is the prediction map for that category. The mIoU is obtained by calculating the average of all category IoUs. This metric is more indicative of the accuracy of the model's classification; therefore, more emphasis will be placed on the model's performance on this metric in the experiments [53]

$$\text{IoU} = \frac{|A \cap B|}{|A \cup B|}. \tag{9}$$

C. Ablation Experiments

Ablation experiments were used to verify the effectiveness of the different components in IAFFNet. The baseline model for this experiment was DeepLab v3+. To ensure reliable results, we conducted multiple trials and averaged the values as presented in Table III. Compared to solely inputting Sentinel-2 imagery, DeepLab v3+ showed improvements of 4.08% and 4.93% in oPA and mIoU, respectively, when SDI was introduced as input. This clearly demonstrates that the inclusion of IK significantly improves the accuracy of the interpretation results.

The experimental results demonstrate the positive effects of the AIF and GFD modules on enhancing the interpretability of the model. In Table III, IAFFNet with AIF achieved higher oPA and mIoU for all three different input conditions compared to the baseline model. AIF significantly enhances the model's learning capabilities for two different features, and its channel attention mechanism facilitates a more effective fusion of these features. With the addition of GFD, IAFFNet shows improvements of 1.11% and 1.91% in oPA and mIoU, respectively, for the SDI input. This indicates that the GFD

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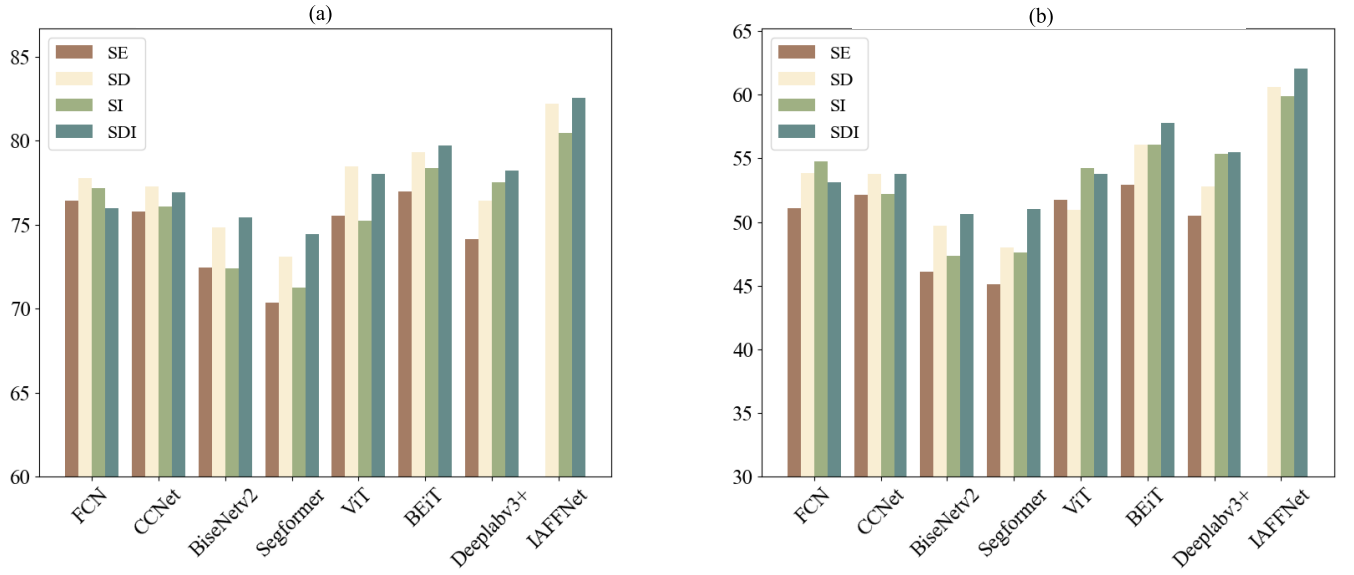


Fig. 12. Comparison between milestone methods and IAFFNet with different inputs. (a) oPA and (b) mIoU of methods with different inputs.

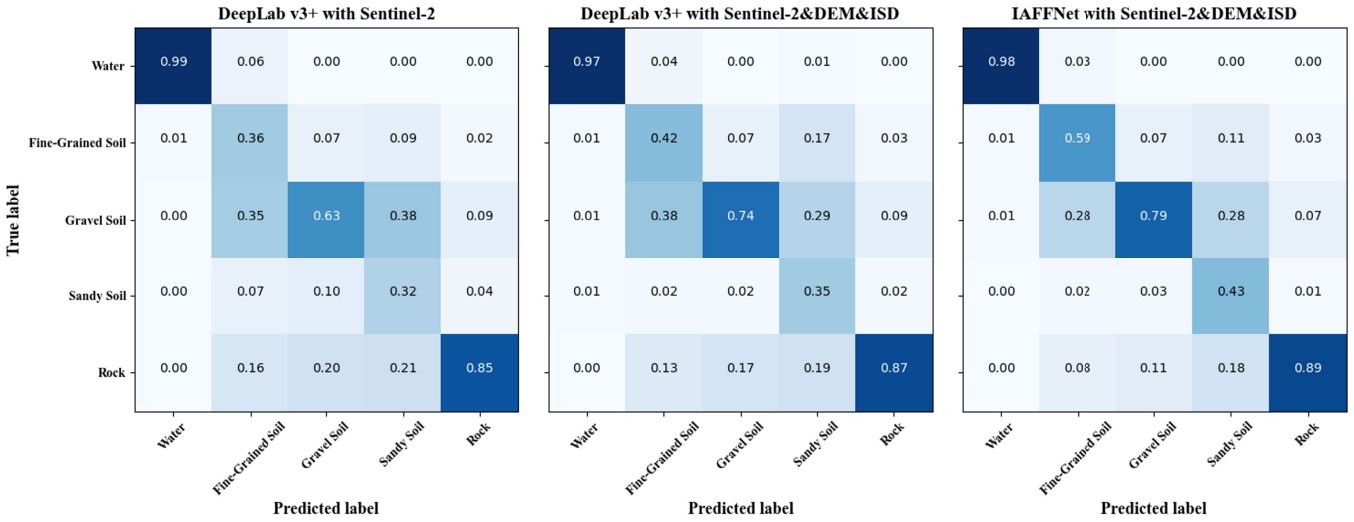


Fig. 13. Confusion matrices of the base model (DeepLab v3+) and our method (IAFFNet).

established by GFD contributes to the model's ability to learn the interdependencies among features, enhancing its learning ability of intrinsic connections within global features. Furthermore, under the same input conditions, IAFFNet consistently outperforms the baseline model in terms of both oPA and mIoU.

D. Comparison Experiments and Analysis

Comparison experiments were conducted to validate the effectiveness and generality of SIFCIM across different models, as well as to investigate how IK influences the interpretability accuracy of each model. Additionally, another objective of the experiments was to demonstrate that IAFFNet outperforms current representative deep-learning models in the task of soil feature interpretation.

Fig. 12 illustrates the overall performance of each model under different input scenarios. All models show varying degrees of improvement in both oPA and mIoU after incorporating IK, which demonstrates the effectiveness and generality of SIFCIM. In most networks, the interpretability accuracy for a double IK input is higher than that for a single IK input. This is because ISD contains horizontal distribution characteristics of soil features, while DEM captures vertical terrain features. The combination of these two IK inputs reveals the spatial distribution patterns of soil features in 3-D space.

However, different models demonstrate varying performance in extracting and integrating useful information from multiple sources. Models such as FCN, CCNet, ViT, and so on exhibit lower oPA when both IK inputs are used compared to a single IK input. This is because these models lack effective fusion of IK data from different sources. IAFFNet, due to the presence of AIF, can extract and utilize valuable information from multiple sources of data more effectively.

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TABLE IV
PA RESULTS OF MILESTONE MODELS AND THE IAFFNET (%)

Methods	Input	WT	FGS	GS	SS	RK
FCN	SE	98.16	38.29	74.22	8.54	84.19
	SD	98.36	37.45	74.42	22.58	86.98
	SI	98.43	35.02	77.60	32.90	82.87
	SDI	98.17	33.99	70.32	33.95	85.88
CCNet	SE	98.14	46.43	66.66	16.10	85.22
	SD	98.75	42.07	70.22	21.98	87.29
	SI	98.42	42.40	71.09	16.99	83.57
	SDI	98.62	43.59	69.92	19.42	86.25
BiseNetv2	SE	98.04	28.51	68.04	3.57	81.97
	SD	98.26	28.86	71.32	15.08	84.94
	SI	98.18	29.94	68.39	9.98	80.51
	SDI	98.18	32.13	73.55	14.96	83.64
Segformer	SE	98.02	31.11	63.82	4.65	78.57
	SD	97.12	35.84	64.97	8.97	83.76
	SI	97.69	32.17	63.40	17.34	80.28
	SDI	98.35	36.10	65.99	23.22	84.98
ViT	SE	98.33	38.21	69.85	18.76	84.42
	SD	98.36	38.32	75.99	18.59	87.62
	SI	98.56	35.23	70.46	17.46	84.07
	SDI	98.50	42.30	71.35	18.06	88.67
BEiT	SE	98.57	39.38	72.50	16.99	85.94
	SD	98.59	42.70	76.49	22.48	87.64
	SI	98.44	40.96	76.82	28.43	85.09
	SDI	98.60	43.82	75.60	31.28	88.44
DeepLab v3+	SE	**98.75**	36.38	63.29	32.04	85.30
	SD	98.59	40.69	71.12	26.10	84.35
	SI	98.23	55.00	70.70	29.08	82.81
	SDI	96.97	42.31	73.95	34.69	86.78
IAFFNet	SD	98.36	59.90	76.63	34.04	**89.00**
	SI	98.06	**61.02**	70.50	37.78	88.50
	SDI	98.46	59.38	**78.05**	**42.46**	88.47

*INPUT: Sentinel-2 (SE), Sentinel-2&DEM (SD), Sentinel-2&ISD (SI), Sentinel-2&DEM&ISD (SDI).

CLASSES: Water (WT), Fine-Grained Soil (FGS), Gravel Soil (GS), Sandy Soil (SS), Rock (RK).

Particularly, IAFFNet shows significantly higher improvement in performance when multiple IK inputs are used, surpassing the performance of other models.

Tables IV and V present the class-wise PA and IoU results for each model, providing a visual representation of the impact of incorporating IK on the interpretation of different soil features and highlighting the advantages of the proposed IAFFNet over other models. For most models, there is an improvement in PA for the majority of the classes after incorporating IK. This demonstrates that IK has a positive impact on the accuracy of soil feature interpretation.

Specifically, when the input is SD, multiple models show significant improvement in PA and IoU for gravel soil, sandy soil, and rock. This is because the terrain distribution features of these three soil elements are more prominent, allowing the models to effectively learn these features from the DEM data.

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TABLE V
IOU RESULTS OF MILESTONE MODELS AND THE IAFFNET (%)

Methods	Input	WT	FGS	GS	SS	RK
FCN	SE	95.82	29.10	53.81	7.29	69.37
	SD	95.18	28.20	55.29	16.96	73.53
	SI	95.71	27.39	54.68	24.64	71.46
	SDI	95.61	25.29	52.15	21.68	70.87
CCNet	SE	95.77	30.51	50.71	12.91	70.66
	SD	95.76	29.89	53.30	17.01	73.06
	SI	95.66	28.47	53.03	13.52	70.47
	SDI	95.84	29.57	53.14	15.89	72.56
BiseNetv2	SE	93.66	21.81	47.30	3.23	63.94
	SD	94.70	22.57	51.60	11.72	67.81
	SI	94.67	22.14	47.45	8.05	64.35
	SDI	95.45	24.39	52.13	11.92	69.29
Segformer	SE	94.01	23.25	43.74	3.61	60.82
	SD	94.14	25.65	48.32	7.31	64.50
	SI	94.44	22.80	44.95	13.26	62.50
	SDI	94.56	27.44	50.10	17.82	65.33
ViT	SE	95.37	30.31	51.49	14.82	66.67
	SD	95.61	30.74	57.75	15.87	71.26
	SI	95.79	28.18	51.64	12.92	66.34
	SDI	95.35	32.78	55.09	13.65	72.03
BEiT	SE	95.81	31.03	54.65	14.59	68.47
	SD	95.78	34.82	58.02	19.71	72.23
	SI	95.88	33.80	57.54	23.90	69.27
	SDI	95.76	34.80	58.75	26.89	72.82
DeepLab v3+	SE	94.79	28.88	49.01	10.75	69.24
	SD	94.69	30.15	54.03	14.86	70.13
	SI	95.69	39.01	54.35	17.14	70.43
	SDI	93.61	32.55	56.62	21.79	72.72
IAFFNet	SD	95.68	44.68	62.71	22.17	**77.87**
	SI	**96.11**	44.72	57.78	26.87	73.85
	SDI	95.98	**45.28**	**62.92**	**28.48**	77.72

*INPUT: Sentinel-2 (SE), Sentinel-2&DEM (SD), Sentinel-2&ISD (SI), Sentinel-2&DEM&ISD (SDI).

CLASSES: Water (WT), Fine-Grained Soil (FGS), Gravel Soil (GS), Sandy Soil (SS), Rock (RK).

In contrast, the terrain distribution features of fine-grained soil are relatively less distinct, resulting in a lack of significant improvement in accuracy in certain models such as FCN, BiSeNet v2, CCNet, and ViT.

When the input is SI, the improvement in models is mainly evident in fine-grained soil, gravel soil, and sandy soil. This is because sandy soil, fine-grained soil, and gravel soil are generally distributed in relation to the distance from water bodies, showing a certain regularity in ISD. This is also the advantage of selecting water bodies as interpretation markers. The distribution of rocks is less influenced by this factor, resulting in a lack of significant improvement in most models for this category.

Lastly, when SDI is used as input, for most models, there is a significant improvement in the interpretation of gravel and sandy soil types. Based on the previous

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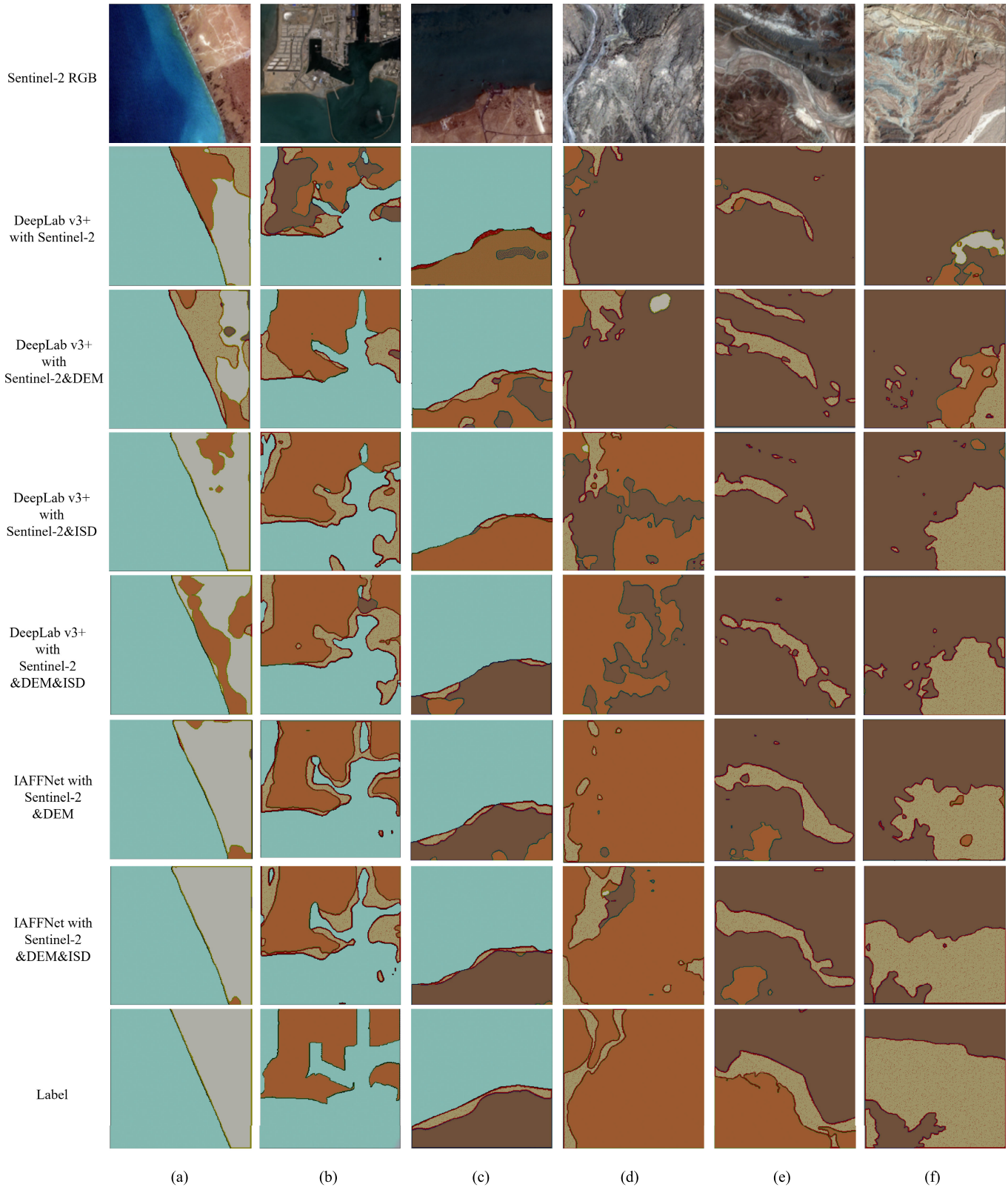


Fig. 14. Part of predicted examples by DeepLab v3+ and IAFFNet. (a) Inclusion of DEM and ISD improves the model's classification of Sandy Soil. (b) Inclusion of IK helps the model identify gravel soil and boundaries. (c) Both IK types improve ability of different methods to identify graval soil. (d) Inclusion of ISD leads to a notable improvement in the prediction of gravel and fine grained soils. (e) Addition of IK improves accuracy of fine-grained soil. (f) Inclusion of IK improves the identification of challenging fine-grained soil.

analysis, these two categories exhibit distinct features in both DEM and ISD. The models can simultaneously learn both IK features, enabling more accurate classification of these soil types. However, models with poor feature fusion capabilities such as FCN, CCNet, and ViT fail to achieve this improvement.

Overall, regardless of the input, IAFFNet consistently achieves higher accuracy in the interpretation of various soil

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features. Particularly, when SDI is used as input, the presence of AIF allows IAFFNet to delve deeper into learning multiple IK features, resulting in improved accuracy for fine-grained soil, gravel soil, and sandy soil.

To further demonstrate how the inclusion of IK makes the different feature categories more distinguishable, we present the confusion matrices in Fig. 13 for three conditions. Before incorporating IK, DeepLab v3+ tends to misclassify fine-grained soil and sandy soil as gravel soil. However, with the inclusion of IK, this misclassification is softened. With the use of IAFFNet, the misclassification of all three soil feature categories is significantly mitigated, indicating that the network makes better use of IK, leading to more effective learning and fusion of features.

Fig. 14 provides a visual representation of the impact of different IK types and combinations on interpretability. In Fig. 14(a), the use of only DEM did not improve the classification accuracy of sandy soil for DeepLab v3+, but with the addition of ISD, this issue is significantly addressed. In Fig. 14(b), the inclusion of IK helps the model identify gravel soil, and IAFFNet performs better than the baseline in handling object boundaries. In Fig. 14(c), when each IK is used individually, the misclassification of rocks does not show significant improvement. However, when both IK types are combined, the baseline's ability to predict rocks is significantly enhanced. In Fig. 14(d), the inclusion of ISD leads to a notable improvement in the prediction of gravel and fine-grained soils. In Fig. 14(e), the addition of IK enhances the classification accuracy of fine-grained soil and highlights IAFFNet's advantage in handling contour details. Finally, in Fig. 14(f), IK improves the identification of challenging fine-grained soil.

It can be observed that the baseline model has a weaker ability to learn from DEM features. The introduction of DEM only leads to a marginal improvement, whereas IAFFNet achieves better results under the same input. This indicates that IAFFNet exhibits robustness to different IK types and combinations, demonstrating high performance, and even achieving results close to the labels in Fig. 14(a), (c), and (f).

The analysis of the above figures and tables leads to the conclusion that the inclusion of IK provides the models with distinctive features of various soil elements, such as terrain and distances to interpretation symbols. This makes it easier to distinguish difficult-to-separate soil types. Consequently, the overall interpretive accuracy is improved, leading to significant enhancements in the final results and showcasing the effectiveness and generality of SIFCIM. Furthermore, IAFFNet's feature fusion capability enables it to perform exceptionally well across different IK types and combinations, reducing the impact of redundant information between different data sources and allowing each data source to contribute its strengths effectively.

E. Complexity Analysis

To evaluate the computational overhead of IAFFNet, two metrics—model parameters and computational complexity (GFLOPs)—were selected for the complexity analysis of DeepLab v3+ and IAFFNet. The specific results are presented in Table VI. The inclusion of IK did not affect the model parameters and GFLOPs of DeepLab v3+, but the number of model parameters and GFLOPs of IAFFNet were approximately 2× and 1.5× higher, respectively, than those of DeepLab v3+ because of the introduction of the two-branch structure and additional modules. This indicates that the computational overhead of IAFFNet was within a reasonable range relative to the performance gains it achieves. It can be seen that the addition of AIF and GFD did not increase the number of IAFFNet parameters too much, and the increase in GFLOPs came more from the use of the two-branch structure.

TABLE VI
PARAMETERS AND COMPUTATIONAL COMPLEXITIES OF DEEPLAB V3+ AND IAFFNET

MODEL	INPUT SIZE	PARAMETERS	Computational Complexities (GFLOPs)
DeepLab v3+	10×224×224	45.20 M	78.83
	10×224×224 & 2×224 ×224	45.20 M	78.84
IAFFNet	10×224×224 & 2×224 ×224	103.91 M	120.76

V. CONCLUSION

In this study, addressing the issue of low accuracy in soil elements interpretation due to inadequate knowledge application and insufficient algorithmic inference and self-learning, we propose the SIFCIM framework and develop IAFFNet based on it. Through extensive experiments and analysis, we have validated the feasibility and generality of utilizing SIFCIM to enhance the accuracy of soil elements interpretation, particularly for soil elements that are difficult to be reflected by remote-sensing imagery, such as sandy soil and gravelly soil. However, the inclusion of IK in this framework can result in redundant information in the data, leading to limited or even decreased accuracy improvements in certain models. Therefore, through empirical evidence, we demonstrate that IAFFNet with an adaptive feature fusion mechanism and GFD mechanism can accurately extract effective features from multisource data, reducing the negative impact of redundant information and fully leveraging the advantages of SIFCIM.

The outcomes of this research may provide inspiration for researchers involved in large-scale intelligent soil element mapping methods and other individuals engaged in knowledge-intensive geological interpretation studies. However, there is still potential for improvement in our work. For instance, as the distance from the soil to the ocean increases, the improvement in interpretation accuracy diminishes with the ISD, necessitating a more scientific quantification approach. Moreover, SIFCIM still has drawbacks such as information redundancy, increased computational complexity, and the need for manual selection of IK. In future research, we will further

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investigate how to address these limitations and propose more scientifically grounded IK quantification algorithms.

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Yue Lu received the B.E. degree in chemical engineering and technology from Nanjing Forestry University, Nanjing, China, in 2020. He is currently pursuing the M.E. degree in computer technology with the China University of Geosciences, Wuhan, China.

His research interests include deep learning, remote-sensing geological interpretation, and multimodal fusion.



Kang He received the master's degree in mining engineering from Central South University, Changsha, China, in 2021. He is currently pursuing the Ph.D. degree in military geology with the China University of Geosciences, Wuhan, China.

His research interests include geological environmental remote sensing and off-road mobility analysis.



Haoran Xu received the B.E. degree in surveying and mapping engineering from the China University of Geosciences, Wuhan, China, in 2021, where he is currently pursuing the Ph.D. degree in geoscience information engineering.

His research interests include unmanned aerial vehicle (UAV) remote sensing, unmanned systems, and remote-sensing data analysis.



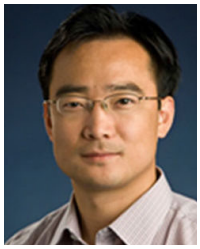
Yusen Dong received the B.S. degree in geology, the M.S. degree in geography, and the Ph.D. degree in cartography and geographic information systems from the China University of Geosciences, Wuhan, China, in 1999, 2002, and 2007, respectively.

From 2006 to 2008, he was engaged in cooperative research in Interferometric Synthetic Aperture Radar (InSAR), Differential InSAR (DInSAR), Permanent Scatterer InSAR (PSInSAR), at the University of New South Wales, Sydney, Australia. He is currently an Associate Professor with the School of Computer, China University of Geosciences. His research interests include DInSAR and ground deformation monitoring, geological remote sensing and remote-sensing investigation of land and resources, and glacier change monitoring.



Wei Han (Member, IEEE) received the B.S. and Ph.D. degrees from the China University of Geosciences, Wuhan, China, in 2015 and 2021, respectively.

He is currently an Associate Professor with the School of Computer Science, China University of Geosciences. His research interests include high-resolution remote-sensing image processing, geological remote sensing, and high-performance computing.



Lizhe Wang (Fellow, IEEE) received the B.E. and M.E. degrees in electrical engineering and automation from Tsinghua University, Beijing, China, in 1998 and 2001, respectively, and the D.E. degree in applied computer science from the University of Karlsruhe, Karlsruhe, Germany, in 2007.

He is currently the Vice-President of the China University of Geosciences, Wuhan, China. His research interests include remote-sensing data processing, digital Earth, and big data computing.

Mr. Wang is a fellow of Institution of Engineering and Technology (IET) and British Computer Society (BCS). He is also an Associate Editor of *Remote Sensing* and *International Journal of Digital Earth*.



Dong Liang received the B.Sc. degree in applied mathematics from the University of Hull, Hull, U.K., in 2006, the M.Sc. degree in applied mathematics from Mälardalen University, Västerås, Sweden, in 2009, and Ph.D. degree in cartography and geographic information system from the University of Chinese Academy of Sciences, Beijing, China, in 2022.

He is currently serving as Assistant Director-General of the International Research Center of Big Data for Sustainable Development Goals (CBAS), Deputy Secretary-General of the Digital Belt and Road Program (DBAR), and Deputy Secretary-General of the Chinese National Committee of the International Society for Digital Earth. His research interests include polar remote sensing and Big Earth Data applications.

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